

SIMERA SENSE

xScape RS422,485

Interface Control Document

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Document History

Revision	Date	Details
1	2021-06-23	First Release
2	2021-07-27	Added UART configuration Updated transaction diagrams to show synchronisation word byte order
3	2021-08-26	Added 16-bit Length field to the Standard protocol definition
4	2024-02-05	Updated section 3.2 – Diagrams now show correct minimum and maximum number of bytes. Updated section 3.3 – Added CCSDS optional 2-byte Ancillary Data field.

Table of Contents

1 Introduction.....	7
1.1 Identification	7
1.2 Intended Use	7
1.3 Context and Summary.....	7
2 Referenced Documents	8
3 Interface Description.....	9
3.1 Physical Interface	10
3.2 Simera Sense Standard Protocol	10
3.2.1 Command Transactions	10
3.2.2 Request Transactions.....	10
3.3 CCSDS Packet Protocol	12
3.3.1 Command Transactions	12
3.3.2 Request Transactions.....	12
3.4 Software Interface.....	16
3.4.1 RS-422/485 Requests	16

List of Figures

Figure 3-1: RS-422/485 Command Transaction – Simera Sense Standard Protocol11

Figure 3-2: RS-422/485 Request Transfer – Simera Sense Standard Protocol11

Figure 3-3: RS-422/485 Request Response Transfer – Simera Sense Standard Protocol11

Figure 3-4: RS-422/485 Command Transaction – CCSDS Packet Protocol13

Figure 3-5: RS-422/485 Request Transfer – CCSDS Packet Protocol14

Figure 3-6: RS-422/485 Request Response Transfer – CCSDS Packet Protocol.....15

List of Tables

Table 2-1: Referenced Documents8

Table 3-1: Standard vs Actual Baud Rates9

Table 3-2 : RS-422/485 Pinout on Connector H210

Table 3-3: Additional Request Identifiers16

List of Abbreviations

Abbreviation	Description
CSK	CubeSat Kit
kbps	Kilobits per second
RX	Receive
TX	Transmit
UART	Universal Asynchronous Receiver-Transmitter

1 Introduction

1.1 Identification

Item Description: xScape RS-422

1.2 Intended Use

This document describes the xScape optional RS-422 UART interface.

1.3 Context and Summary

The xScape range of products support various options for control and/or data interfaces, including RS-422.

The RS-422 interface provides a control interface only.

2 Referenced Documents

Table 2-1 lists documents that are referenced in this document. In the event of conflict between the contents of the reference documents and this document, this document shall take precedence.

Table 2-1: Referenced Documents

Ref. #	Reference
[1]	CCSDS 133.0.B-1 - Space Packet Protocol, Recommended Standard
[2]	051715 - xScape Control and Data Interface Control Document

For undated references, the latest released version of the reference document applies. For dated references, subsequent versions of the document do not apply. It is best practice to always refer to the latest released version. Unless otherwise stated, web links referenced above were last accessed at the release date of the current version of this document.

3 Interface Description

The imager provides a UART interface at either RS-422 or RS-485 levels that is routed to the CSK bus compatible connector H2. The UART configuration is “8N1”: eight-bit characters, with one start bit, one stop bit, and no parity bit. This interface can operate at a fixed baud rate ranging from approximately 9.6 kbps up to 460.8 kbps. The baud rate is not configurable via the Imager control interface but is provided as a build option and is programmed before shipping. The specific baud rates supported are calculated as follows:

$$\text{Baud Rate} = \frac{100\,000\,000}{(\text{value} + 1) \times 16}$$

where *value* may be between 13 and 650. Hardware which interfaces to the imager should be configured with a matching baud rate (with an error of less than 2%). Table 3-1 shows actual programmed baud rates including error as a percentage for some standard baud rates.

Table 3-1: Standard vs Actual Baud Rates

Baud	<i>value</i>	Actual Baud	Error
460800	13	446428.571	3.12%
230400	26	231481.481	0.47%
115200	53	115740.741	0.47%
76800	80	77160.494	0.47%
57600	108	57339.450	0.45%
38400	162	38343.558	0.15%
28800	216	28801.843	0.01%
19200	325	19171.779	0.15%
9600	650	9600.614	0.01%

Two standard protocol options are available. The Simera Sense Standard Protocol follows a minimalistic approach, while the CCSDS packet protocol follows the recommendation as per [1]. The CCSDS packet protocol is available with and without Secondary Header option. The protocol cannot be changed on-the-fly and is a build option.

3.1 Physical Interface

The RS-422/485 signals are routed to the CSK interface connector H2 pins as shown in Table 3-2.

Table 3-2 : RS-422/485 Pinout on Connector H2

Pin	Signal	Description
47	RS422_RX_A	RS-422 Receiver Line "A"
49	RS422_RX_B	RS-422 Receiver Line "B"
48	RS422_TX_A	RS-422 Transmitter Line "A"
50	RS422_TX_B	RS-422 Transmitter Line "B"
47 or 48	RS485_A	RS-485 Transceiver Line "A"
49 or 50	RS485_B	RS-485 Transceiver Line "B"

3.2 Simera Sense Standard Protocol

For both command and request transactions, the RS-422/485 transfers consist of a synchronisation word followed by a 16-bit length field (in Little Endian) and the Imager standard command or request variable length packet (refer to [2]) in little endian format. The value of the 16-bit Length field equals the number of bytes that follow, i.e., the length of the packet in bytes.

3.2.1 Command Transactions

Command transactions require the RS-422/485 host to send a command transfer, as shown in Figure 3-1.

3.2.2 Request Transactions

Request transactions are carried out by two transfers. Firstly, the RS-422/485 host to send a request transfer, as shown in Figure 3-2. The Imager then responds by sending a request response transfer, as shown in Figure 3-3, to the host.

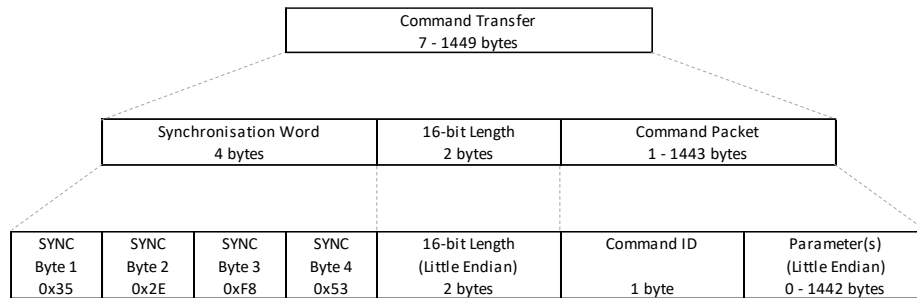


Figure 3-1: RS-422/485 Command Transaction – Simera Sense Standard Protocol

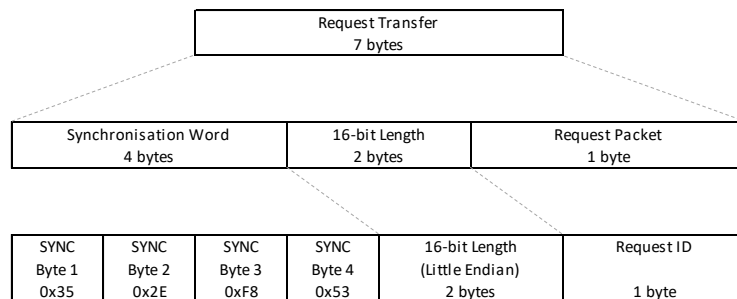


Figure 3-2: RS-422/485 Request Transfer – Simera Sense Standard Protocol

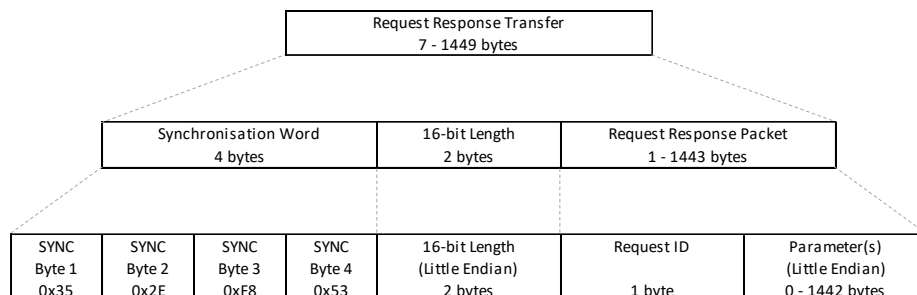


Figure 3-3: RS-422/485 Request Response Transfer – Simera Sense Standard Protocol

3.3 CCSDS Packet Protocol

For both command and request transactions, the RS-422/485 transfers consist of a synchronisation word followed by a variable length CCSDS packet. The CCSDS packet's primary and secondary headers are all represented in big endian format. The Imager standard command or request packet (refer to [2]) is inserted into the CCSDS packet data field in its original, little endian, format. The CCSDS packet secondary header is optional and consists of a 9-byte Time Code field and an optional 2-byte Ancillary Data field.

3.3.1 Command Transactions

Command transactions require the RS-422/485 host to send a command transfer to the imager as shown in Figure 3-4. Fields marked "N/A" are not applicable (don't care) to the transaction and their contents are ignored.

3.3.2 Request Transactions

Request transactions are carried out by two transfers. Firstly, the RS-422/485 host to send a request transfer, as shown in Figure 3-5. Fields marked "N/A" are not applicable (don't care) to the transaction. The Imager then responds by sending a request response transfer, as shown in Figure 3-6, to the host. Fields marked "N/A" are not applicable (don't care) to the transaction. The value for the request response transfer APID field can be specified and is a build option. If enabled, the value for the Secondary Header Ancillary Data field can also be specified as a build option.

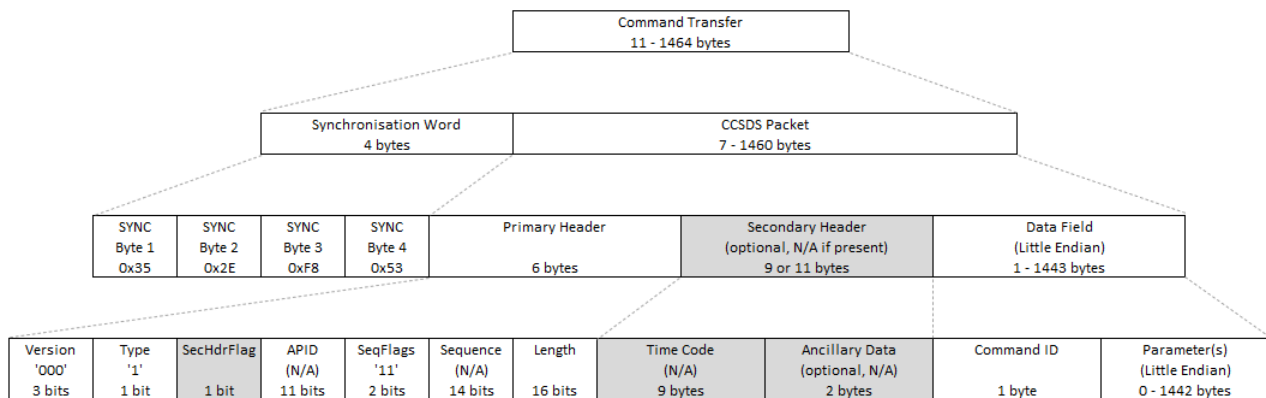


Figure 3-4: RS-422/485 Command Transaction – CCSDS Packet Protocol

Notes:

- When the Secondary Header Flag ("SecHdrFlag") is set, it signifies the presence of the Secondary Header. When present, the contents of the Secondary Header is not used by the Imager and can be set to any value.
- Primary Header "APID" field is not used by the Imager and can be set to any value.
- Primary Header "Sequence" field is not used by the Imager and can be set to any value.
- The Primary Header "Length" is set to the number of remaining bytes in the transfer minus 1 byte, as per [1].

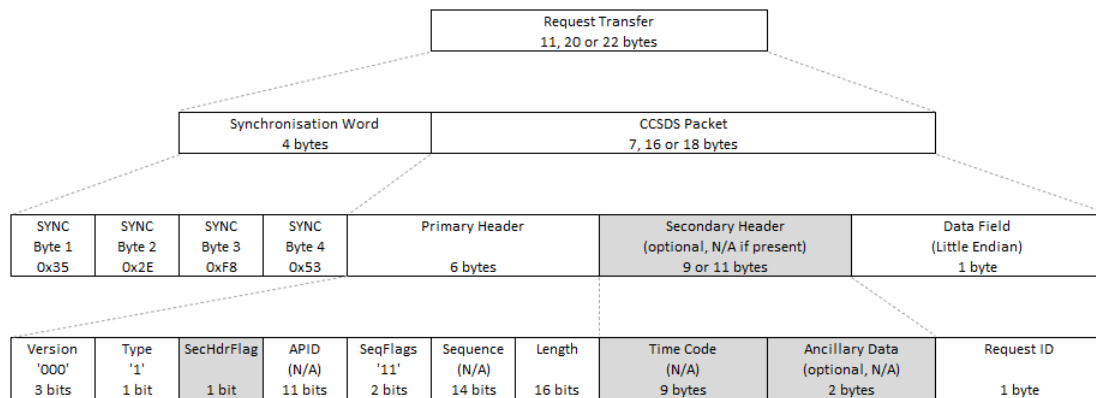


Figure 3-5: RS-422/485 Request Transfer – CCSDS Packet Protocol

Notes:

- When the Secondary Header Flag ("SecHdrFlag") is set, it signifies the presence of the Secondary Header. When present, the contents of the Secondary Header is not used by the Imager and can be set to any value.
- Primary Header "APID" field is not used by the Imager and can be set to any value.
- Primary Header "Sequence" field is not used by the Imager and can be set to any value.
- The Primary Header "Length" is set to the number of remaining bytes in the transfer minus 1 byte, as per [1].

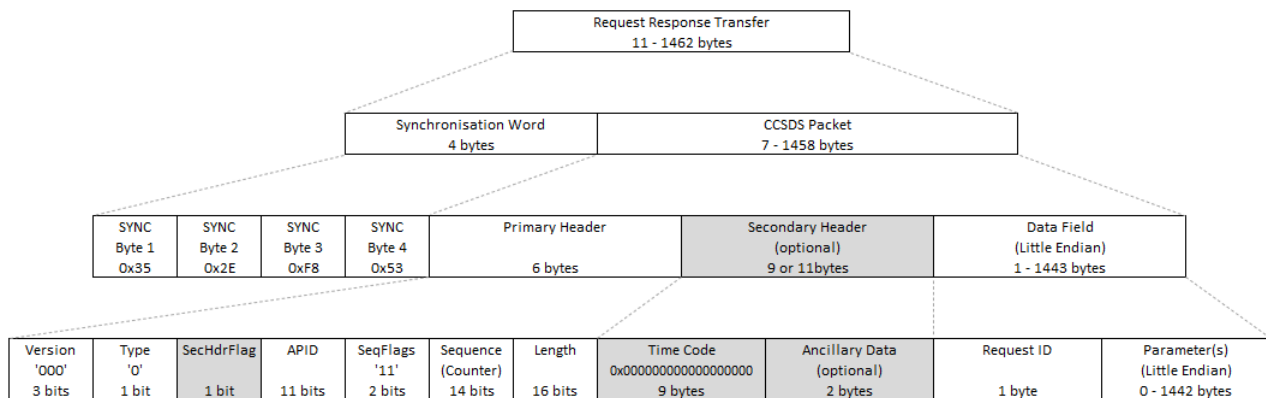


Figure 3-6: RS-422/485 Request Response Transfer – CCSDS Packet Protocol

Notes:

- When the Secondary Header Flag ("SecHdrFlag") is set, it signifies the presence of the Secondary Header. When present, the contents of the Time Code field is 0x0000000000000000. The Ancillary Data field can optionally be added to the Secondary Header and can be set to any 16-bit value as a build option.
- Primary Header "APID" field is not used by the Imager and defaults to 0x000. It can however be set to any value (a fixed value, the same for all request response packets) if desired as a build option.
- Primary Header "Sequence" field is incremented for each packet transmitted. Its value is reset to 0x0000 at Imager reset or power cycle.
- The 16-bit Primary Header "Length" field is set to the number of remaining bytes in the transfer minus 1 byte, as per [1].

3.4 Software Interface

3.4.1 RS-422/485 Requests

The inclusion of the RS-422/485 Interface adds the following two Requests.

Table 3-3: Additional Request Identifiers

Request ID	Packet Name	Description
0x97	RS-422/485 UART DIAGNOSTICS	RS-422/485 UART Information
0x98	RS-422/485 PACKET DIAGNOSTICS	RS-422/485 Control Interface Packet Information

3.4.1.1 RS-422/485 UART Diagnostics

The RS-422/485 UART DIAGNOSTICS request returns low-level diagnostic information about the Universal Asynchronous Receive Transmitter peripheral, used by the RS-422/485 Control Interface. This information is used to diagnose communication problems on the RS-422/485 Control Interface.

NOTICE Under normal operating conditions, it will not be necessary to issue this request. The information returned should be shared with the manufacturer only to diagnose RS-422/485 communication problems.

RS-422/485 UART Diagnostics												
Request ID	0x97											
Data Bytes	13											
Return Data	Type	Description										
RX Byte Count	uint32	A count of the total bytes written to the internal buffer since the last reset or power-up.										
RX Byte Used Count	uint32	A count of the total bytes read from the internal buffer since the last reset or power-up.										
RX Byte Dropped	uint32	A count of the bytes dropped due to a buffer overflow.										
RX Error Flags	uint8	Accumulated UART receive errors since this request was last issued.										
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Parity Error</td></tr><tr><td>1</td><td>Overflow Error</td></tr><tr><td>2</td><td>Frame Error</td></tr><tr><td>3..7</td><td>Reserved</td></tr></table>	Bit	Description	0	Parity Error	1	Overflow Error	2	Frame Error	3..7	Reserved
		Bit	Description									
		0	Parity Error									
		1	Overflow Error									
		2	Frame Error									
3..7	Reserved											

3.4.1.2 RS-422/485 Packet Diagnostics

The RS-422/485 PACKET DIAGNOSTICS request returns diagnostic information about the RS-422/485 Packets of the Control Interface only. This information is used to diagnose connection problems on the RS-422/485 Control Interface.

RS-422/485 Packet Diagnostics		
Request ID	0x98	
Data Bytes	17	
Return Data	Type	Description
Packet RX Count	uint32	A count of the total of packets received by the imager since the last reset or power-up.
Packet RX Fail	uint32	A count of the failed packets received by the imager since the last reset or power-up. Failed Packets will have an error, as indicated in the Error Flags field.
Packet RX OK	uint32	A count of the successful packets (no errors) received by the imager since the last reset or power-up.
Packet TX	uint32	A count of the total of packets transmitted by the imager since the last reset or power-up.
Packet RX Error Flags	uint8	Accumulated errors for all packets received since this request was last issued.
		Note: Packet Error Flags are cleared when this request is received.

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